DETERMINISTIC VISUAL COMPARISON

Megalodon From Below — The Attack You Do Not See Coming

The owner's shot, staged in the live game: a diver on the surface who can see nothing, then dives, looks down, and finds a megalodon coming up at him out of the dark. One checkout, one seed, one water column; the BEFORE column sets cfg_SHARK_ASCENT=0 and runs the shipped code path, in which a shark's depth is a constant off the WATERLINE that never once asks where its quarry is — so the pass finishes with the animal ABOVE the diver. The AFTER column measures the staging depth from the quarry and turns the rush into a climb solved every frame to arrive with its own charge.

5 MATCHED SUBJECTS	**1** DEVICE FRAMES	**10** BROWSER SCREENSHOTS

Frames: 1280x720

BEFORE · cfg_SHARK_ASCENT=0 (same build)

`http://127.0.0.1:8726/?seed=90210&cfg_BOOT_METER=0&cfg_SHARK_ASCENT=0&visualCompare=before-1788084905970`

AFTER · §THE ASCENT

`http://127.0.0.1:8726/?seed=90210&cfg_BOOT_METER=0&visualCompare=after-1788085137553`

Generated August 30, 2026 at 10:22 AM. Both sides boot index.html on the same seed, click through to free play, freeze the frame loop and advance the world only through CBZ.stepSim, so a beat is a number of GAME seconds rather than rasterised frames. The coast is found by bisecting the live shore field (never a typed coordinate) and the diver is put in the water at an authored depth through CBZ.citySwimBegin. A megalodon is spawned in the dark beneath him, pointed at him — the bow-on attitude a shark already has by the time its circle becomes a rush — and handed to the production FSM with CBZ.predatorCommit; everything after that is predatorHunt driving the animal. marine_predation is off on BOTH columns, because with it on the megalodon is given a fish four hundred metres away and the hunt under test never runs. Every number is read from the engine's own seams (CBZ.sharkAscentRead, CBZ.sharkAscentAudit) at the instant of the capture, and the body's pitch is derived from its own velocity rather than authored.

Σ

Measurements · 1/3

Read off the engine's own measurement seams at the instant of each capture, on both sides, with identical staging and identical simulated seconds. WHY `belowQuarryM` CARRIES NO DIRECTION. It is the same measurement at two opposite moments and it wants opposite things at each. While the animal is STALKING, deeper under you is better — that is the ambush, and it is what `stagedBelowM` scores. At the moment it ARRIVES, the number should be near ZERO, because the whole point is that it came up to your depth instead of biting you from the dark; that is what `arrivalVerticalM` scores, and marking the raw number `higher is better` made a megalodon closing from 6.89 m under the diver to 3.96 m read as a REGRESSION. One ambiguous metric was a modelling mistake in this preset, not a finding. What the pre-ascent code could never do is either of them on purpose: its depth is a constant off the WATERLINE that never once consults the quarry, so against a diver it drifts and ends the pass ABOVE him — between you and the light, which is exactly backwards.

SUBJECT	METRIC	BEFORE	AFTER	Δ
1 · At The Surface, Looking Around · Nothing	Diver's depth at capture m	1.47	1.43	-3%
1 · At The Surface, Looking Around · Nothing	Camera forward Y (−1 = straight down), calibrated not assumed	-0.05	-0.05	0%
1 · At The Surface, Looking Around · Nothing	Diver's depth (engine read) m	1.02	1	-2%
1 · At The Surface, Looking Around · Nothing	Megalodon's depth m	21.8	24.4	+12%
1 · At The Surface, Looking Around · Nothing	Megalodon BELOW the diver (>0 = underneath). Directionless on purpose — see note m	21	23.6	+12%
1 · At The Surface, Looking Around · Nothing	Horizontal gap to the diver m	10.3	10.3	0%
1 · At The Surface, Looking Around · Nothing, 5 · Where The Attack Actually Ends Up	A solved ascent is running right now	0	0	0
1 · At The Surface, Looking Around · Nothing, 5 · Where The Attack Actually Ends Up	Vertical speed at capture m/s	0	0	0
1 · At The Surface, Looking Around · Nothing, 5 · Where The Attack Actually Ends Up	Nose-up angle while climbing (derived from velocity) °	0	0	0
All 5 matched captures	predatorHunt state at capture	—	—	—
1 · At The Surface, Looking Around · Nothing	Solved ascents begun	0	0	0
1 · At The Surface, Looking Around · Nothing, 2 · Dive, And Look Down, 3 · The Ascent · Six Frames Of A Solved Climb	Height of the last climb m	0	0	0
All 5 matched captures	predatorCommit took (a staging check, not a claim)	1	1	0%
1 · At The Surface, Looking Around · Nothing	Camera depth (what the underwater grade is a function of) m	1.01	0.9	-11%
All 5 matched captures	Underwater fog far plane at capture m	34.7	34.7	0%
All 5 matched captures	Water column under the eye (staging check — must match) m	62	62	0%
1 · At The Surface, Looking Around · Nothing	How deep the megalodon is while you see nothing m	21.8	24.4	+12%
1 · At The Surface, Looking Around · Nothing	STALKING: how far under the diver it stages (deeper = the ambush) m	21	23.6	+12%
2 · Dive, And Look Down	Diver's depth at capture m	10.3	10.3	0%
2 · Dive, And Look Down	Diver's depth (engine read) m	9.99	10	+1%
2 · Dive, And Look Down	Megalodon's depth m	25.2	25.6	+1%
2 · Dive, And Look Down	Megalodon BELOW the diver (>0 = underneath). Directionless on purpose — see note m	15.2	15.3	+1%
2 · Dive, And Look Down	Horizontal gap to the diver m	19.4	19.4	+0%
2 · Dive, And Look Down, 3 · The Ascent · Six Frames Of A Solved Climb, 4 · The Reference Photograph · Jaws, From Underneath	A solved ascent is running right now	0	1	+1

Σ

Measurements · 2/3

Read off the engine's own measurement seams at the instant of each capture, on both sides, with identical staging and identical simulated seconds. WHY `belowQuarryM` CARRIES NO DIRECTION. It is the same measurement at two opposite moments and it wants opposite things at each. While the animal is STALKING, deeper under you is better — that is the ambush, and it is what `stagedBelowM` scores. At the moment it ARRIVES, the number should be near ZERO, because the whole point is that it came up to your depth instead of biting you from the dark; that is what `arrivalVerticalM` scores, and marking the raw number `higher is better` made a megalodon closing from 6.89 m under the diver to 3.96 m read as a REGRESSION. One ambiguous metric was a modelling mistake in this preset, not a finding. What the pre-ascent code could never do is either of them on purpose: its depth is a constant off the WATERLINE that never once consults the quarry, so against a diver it drifts and ends the pass ABOVE him — between you and the light, which is exactly backwards.

SUBJECT	METRIC	BEFORE	AFTER	Δ
2 · Dive, And Look Down	Vertical speed at capture m/s	0	6.71	+6.71
2 · Dive, And Look Down	Nose-up angle while climbing (derived from velocity) °	0	19.9	+19.9
2 · Dive, And Look Down	Solved ascents begun	0	1	+1
2 · Dive, And Look Down	Camera depth (what the underwater grade is a function of) m	8.81	8.86	+1%
2 · Dive, And Look Down	Camera forward Y (−1 = straight down), calibrated not assumed	-0.81	-0.81	0%
3 · The Ascent · Six Frames Of A Solved Climb	Diver's depth at capture m	10.3	10.3	0%
3 · The Ascent · Six Frames Of A Solved Climb	The climb started at all	0	1	+1
3 · The Ascent · Six Frames Of A Solved Climb	STALKING: how far under the diver it stages (deeper = the ambush) m	18.1	20.3	+12%
3 · The Ascent · Six Frames Of A Solved Climb	Diver's depth (engine read) m	10	9.78	-2%
3 · The Ascent · Six Frames Of A Solved Climb	Megalodon's depth m	27.1	29.6	+9%
3 · The Ascent · Six Frames Of A Solved Climb	Megalodon BELOW the diver (>0 = underneath). Directionless on purpose — see note m	17	19.8	+17%
3 · The Ascent · Six Frames Of A Solved Climb	Horizontal gap to the diver m	45.1	45.2	+0%
3 · The Ascent · Six Frames Of A Solved Climb	Vertical speed at capture m/s	0	5.1	+5.1
3 · The Ascent · Six Frames Of A Solved Climb	Nose-up angle while climbing (derived from velocity) °	0	27.2	+27.2
3 · The Ascent · Six Frames Of A Solved Climb	Solved ascents begun	0	2	+2
3 · The Ascent · Six Frames Of A Solved Climb	Camera depth (what the underwater grade is a function of) m	9.01	8.78	-3%
4 · The Reference Photograph · Jaws, From Underneath	Diver's depth at capture m	10.2	10.2	0%
4 · The Reference Photograph · Jaws, From Underneath	Diver's depth (engine read) m	10.4	10.2	-2%
4 · The Reference Photograph · Jaws, From Underneath	Megalodon's depth m	18.7	19.2	+2%
4 · The Reference Photograph · Jaws, From Underneath	Megalodon BELOW the diver (>0 = underneath). Directionless on purpose — see note m	8.79	8.94	+2%
4 · The Reference Photograph · Jaws, From Underneath	Horizontal gap to the diver m	11	11	-1%
4 · The Reference Photograph · Jaws, From Underneath	Vertical speed at capture m/s	0	11.3	+11.3
4 · The Reference Photograph · Jaws, From Underneath	Nose-up angle while climbing (derived from velocity) °	0	28.6	+28.6
4 · The Reference Photograph · Jaws, From Underneath	Solved ascents begun	0	3	+3

Σ

Measurements · 3/3

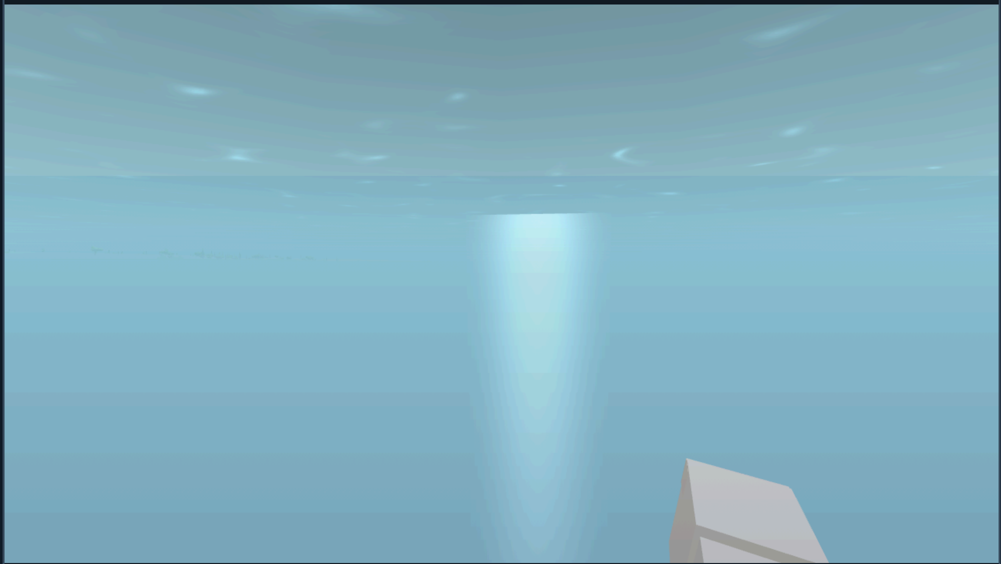
Read off the engine's own measurement seams at the instant of each capture, on both sides, with identical staging and identical simulated seconds. WHY `belowQuarryM` CARRIES NO DIRECTION. It is the same measurement at two opposite moments and it wants opposite things at each. While the animal is STALKING, deeper under you is better — that is the ambush, and it is what `stagedBelowM` scores. At the moment it ARRIVES, the number should be near ZERO, because the whole point is that it came up to your depth instead of biting you from the dark; that is what `arrivalVerticalM` scores, and marking the raw number `higher is better` made a megalodon closing from 6.89 m under the diver to 3.96 m read as a REGRESSION. One ambiguous metric was a modelling mistake in this preset, not a finding. What the pre-ascent code could never do is either of them on purpose: its depth is a constant off the WATERLINE that never once consults the quarry, so against a diver it drifts and ends the pass ABOVE him — between you and the light, which is exactly backwards.

SUBJECT	METRIC	BEFORE	AFTER	Δ
4 · The Reference Photograph · Jaws, From Underneath	Height of the last climb m	0	20.9	+20.9
4 · The Reference Photograph · Jaws, From Underneath	Camera depth (what the underwater grade is a function of) m	10.2	10	-2%
4 · The Reference Photograph · Jaws, From Underneath	Closest it came horizontally (capture-condition readout — this feature is about the VERTICAL) m	12.9	12.8	-1%
4 · The Reference Photograph · Jaws, From Underneath	The animal has an authored mouth to open	1	1	0%
5 · Where The Attack Actually Ends Up	Diver's depth at capture m	10.2	10.2	0%
5 · Where The Attack Actually Ends Up	Diver's depth (engine read) m	9.65	9.48	-2%
5 · Where The Attack Actually Ends Up	Megalodon's depth m	16.5	13.9	-16%
5 · Where The Attack Actually Ends Up	Megalodon BELOW the diver (>0 = underneath). Directionless on purpose — see note m	7.02	3.84	-45%
5 · Where The Attack Actually Ends Up	Horizontal gap to the diver m	1.35	1.42	+5%
5 · Where The Attack Actually Ends Up	Solved ascents begun	0	4	+4
5 · Where The Attack Actually Ends Up	Height of the last climb m	0	18.8	+18.8
5 · Where The Attack Actually Ends Up	Camera depth (what the underwater grade is a function of) m	10.1	9.88	-2%
5 · Where The Attack Actually Ends Up	ARRIVING: vertical gap left when the teeth get there (0 = it came to your depth) m	7.02	3.84	-45%
5 · Where The Attack Actually Ends Up	Closest vertical approach over the pass (0 = it reached your depth) m	7.29	3.53	-52%
5 · Where The Attack Actually Ends Up	Deepest it ever sat under the diver (staging readout — both columns stage the same start depth) m	22	21.9	-1%
5 · Where The Attack Actually Ends Up	Peak climb rate over the whole pass m/s	0	12.6	+12.6
5 · Where The Attack Actually Ends Up	Peak nose-up angle during the climb °	0	34.6	+34.6

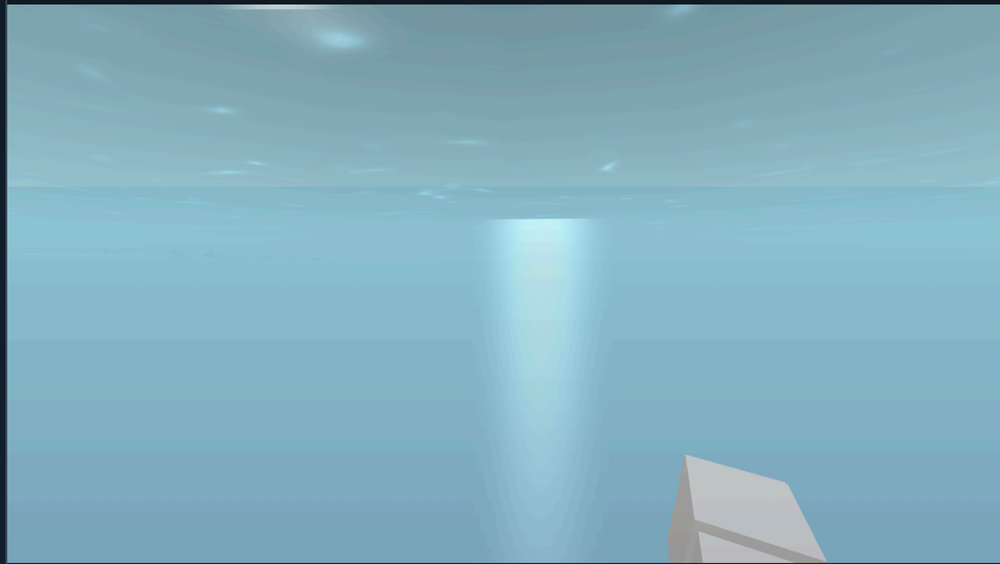
01 1 · At The Surface, Looking Around · Nothing

The diver on the surface, turning to look at the sea around him. This frame is supposed to be EMPTY and it is empty on both sides — that is the point of including it. The megalodon is already there, already hunting him, forty metres down. What the two columns disagree about is not what you can see here; it is what is underneath you while you see nothing.

BEFORE — BEFORE · cfg_SHARK_ASCENT=0 (same build)



AFTER — AFTER · §THE ASCENT



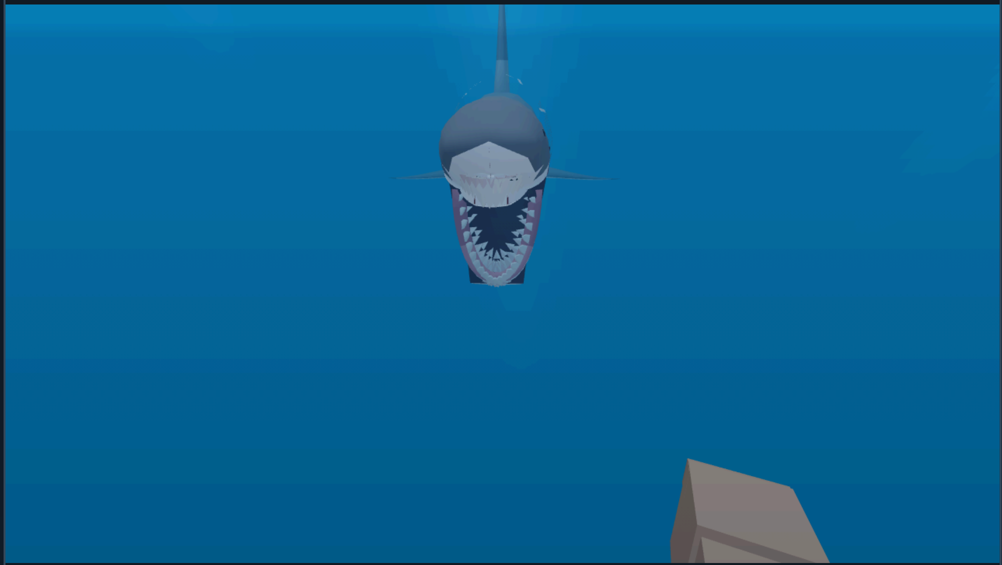
Same checkout · same seed · same water column · same diver · same megalodon · same simulated seconds

The same diver, fifteen metres under, lens pointed at the seabed. BEFORE: the megalodon's depth is a constant off the WATERLINE and takes no notice of the diver at all, so it is level with him or above him — you are looking down at empty water while the animal is between you and the light. AFTER: its staging depth is measured from the quarry, so it is where an ambush predator belongs — underneath, in the dark, looking up at your silhouette.

02 2 · Dive, And Look Down

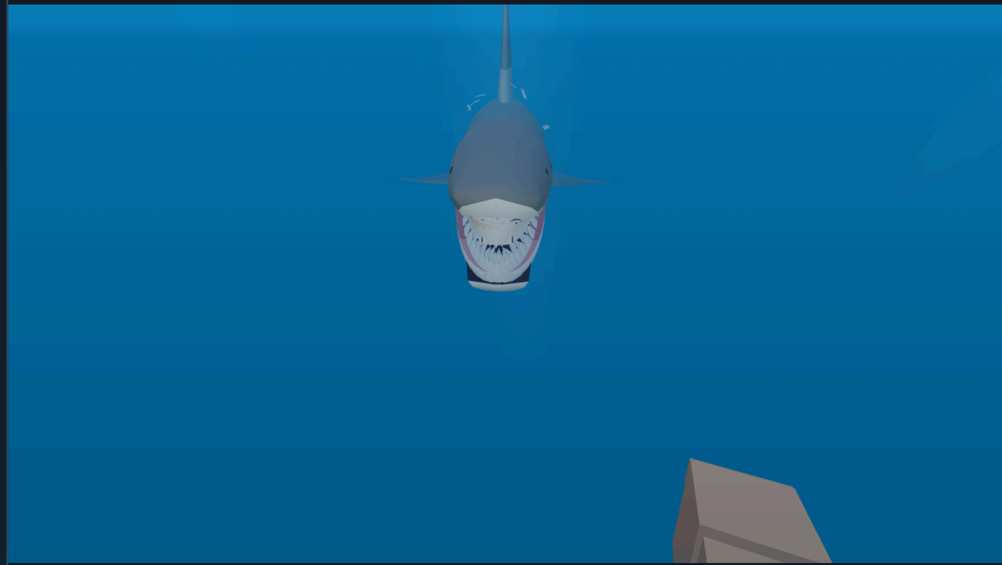
BEFORE

BEFORE · cfg_SHARK_ASCENT=0 (same build)



AFTER

AFTER · §THE ASCENT



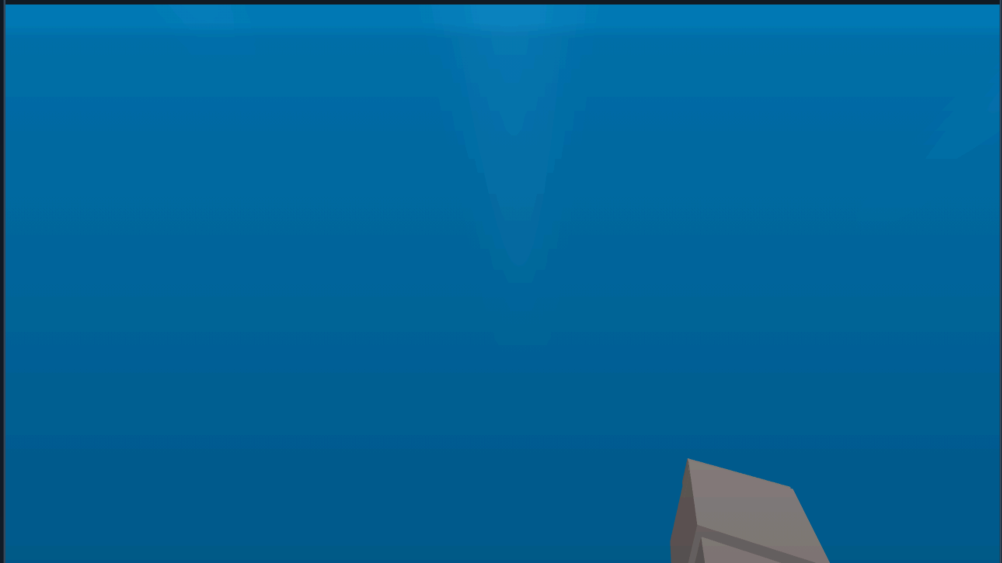
Same checkout · same seed · same water column · same diver · same megalodon · same simulated seconds

03

3 · The Ascent · Six Frames Of A Solved Climb

1.3 s of the same rush on both sides, same inputs, same seconds. BEFORE: there is no climb to film — the rush is a torpedo run at a fixed depth off the waterline and the strip shows the animal holding its level. AFTER: an acceleration solved every frame from the climb remaining, the gap remaining and the speed it is actually making, so it arrives with its own charge. The nose angle is not animated: it is atan2(vertical speed, horizontal speed), the same expression the ballistic breach uses.

BEFORE BEFORE · cfg_SHARK_ASCENT=0 (same build)

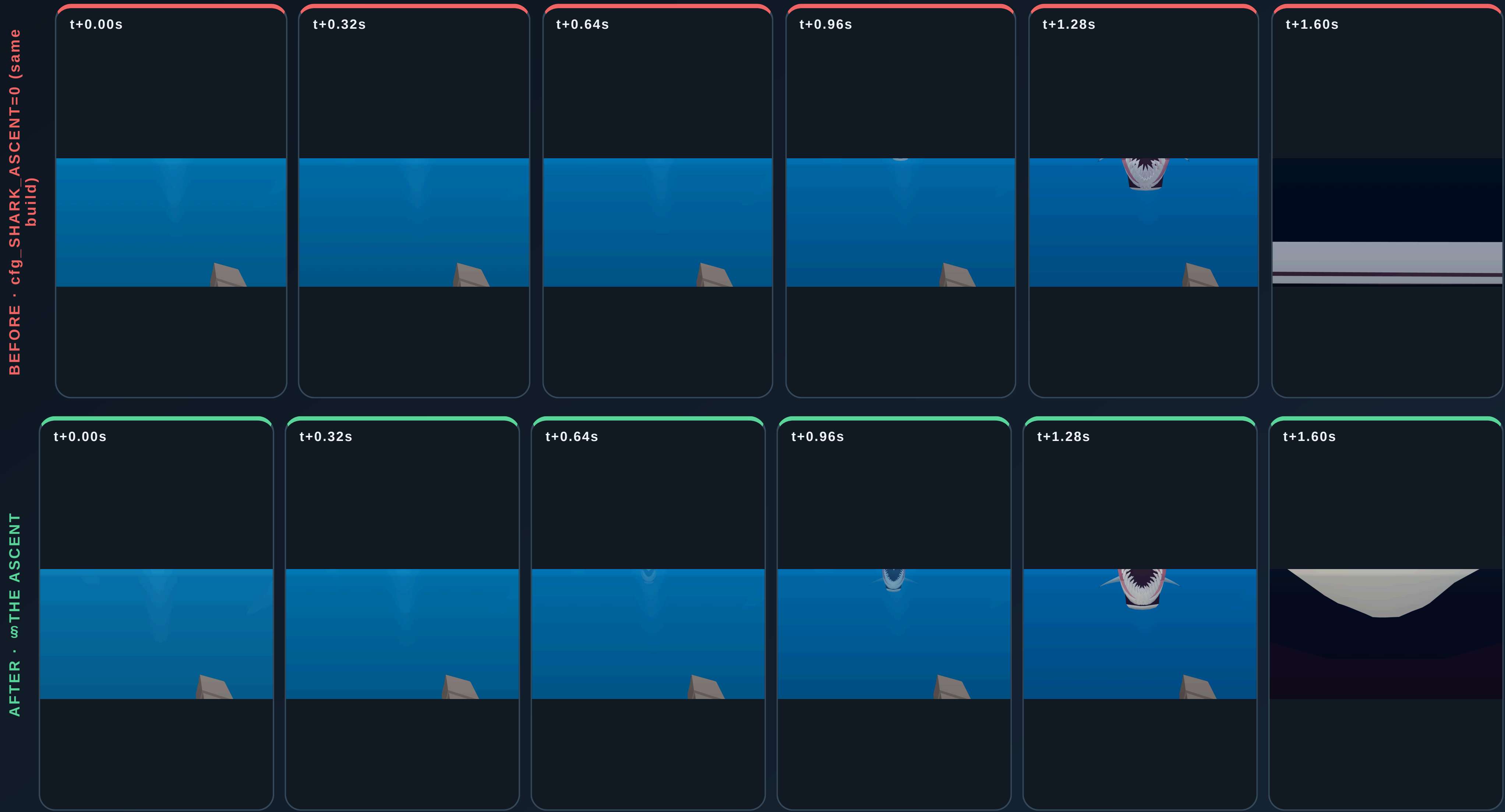


AFTER AFTER · §THE ASCENT

Same checkout · same seed · same water column · same diver · same megalodon · same simulated seconds

3 · The Ascent · Six Frames Of A Solved Climb · over time

The same simulated seconds on both builds, photographed every 0.32s. Motion is the claim; the strip is the proof.



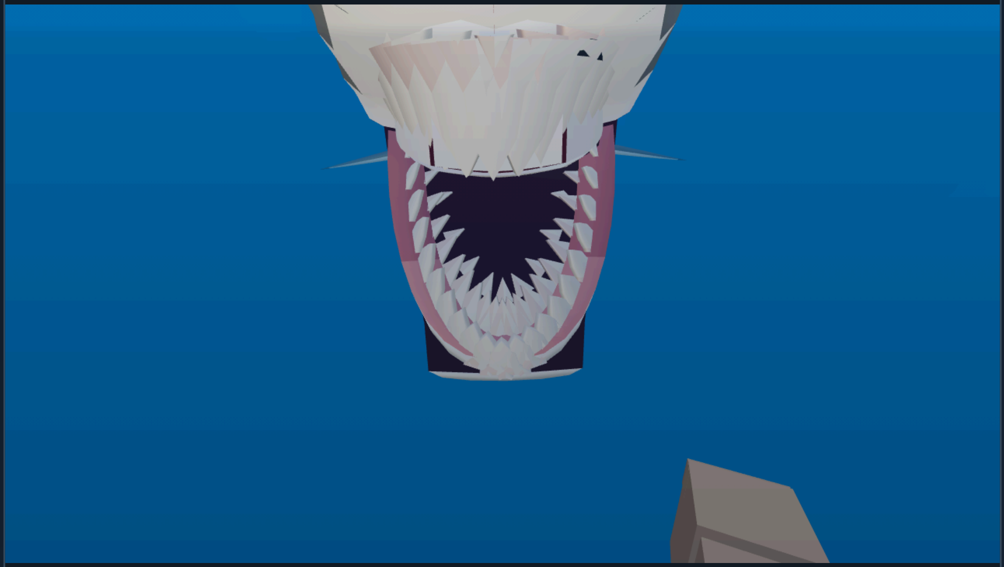
Same checkout · same seed · same water column · same diver · same megalodon · same simulated seconds

The owner's picture: the prey's-eye view, mouth open, filling the frame from below. Both columns show a gape here and that is the honest result — the old code bites you perfectly well, it simply never comes UP to do it, so read the ATTITUDE rather than the mouth. BEFORE: level, the head square-on, arriving out of the horizontal. AFTER: nose-up through the whole approach, the upper jaw protruded and the underside of the head presented, because the body is climbing (climbMS 11.6 vs 0.0, pitchDeg 28.6 vs 0.0 in the table). The gape itself is written ON the climb — docs/SHARK-REFERENCE.md §1, the gape IS the photograph — so the teeth arrive with the animal rather than after it.

04 4 · The Reference Photograph · Jaws, From Underneath

BEFORE BEFORE · cfg_SHARK_ASCENT=0 (same build)

AFTER AFTER · §THE ASCENT



prey-eye · 1280x720

Same checkout · same seed · same water column · same diver · same megalodon · same simulated seconds

05 5 · Where The Attack Actually Ends Up

The last beat, and the numbers that decide whether any of this was real. Both columns are stepped the same fixed 2.2 s from the same commit, and the extrema are tracked across it. BEFORE: no climb at any point (peak 0.0 m/s), and the animal ends the pass ABOVE the diver — it drifted up past him because its depth was never about him in the first place. AFTER: a solved climb out of the dark, arriving at the diver's own depth and HOLDING there while the bite lands, instead of being pulled straight back down to its staging depth mid-strike.

BEFORE

BEFORE · cfg_SHARK_ASCENT=0 (same build)

AFTER

AFTER · §THE ASCENT

Same checkout · same seed · same water column · same diver · same megalodon · same simulated seconds